The Complete Treatise on Econopsychology

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Abstract

This paper outlines Econopsychology as a framework for economic choice under preferences, ambiguity, resource constraints, and a zero inflation risk premium (ZIRP). Exact ZIRP is the theoretical benchmark, distinguished from a policy implementation path permitting temporary deviations within a shrinking envelope. A Gaussian aggregate-return function, a resource-allocation identity, and a nonlinear investment recurrence provide the financial and dynamical starting point. A constrained choice problem connects consumption, investment, and ambiguity. Mathematical consequences are separated from modelling hypotheses and open empirical questions. The purpose is a concise research programme, not a claim that every proposed mechanism has already been established.

The paper ends with "The End"

1 Introduction

Economic choice allocates scarce resources between current consumption and investment intended to generate future benefits. Preferences determine how outcomes are valued; risk and ambiguity affect how uncertain outcomes are assessed; institutions shape the feasible set [3, 5, 6, 7]. Econopsychology is proposed as the study of how preferences and ambiguity interact with economic constraints to shape consumption, investment, and welfare.

The governing principle is that investment is a means to future benefits, not an end in itself. Thus investment accumulation must be distinguished from investment performance and welfare. The financial benchmark is the Zero Inflation Risk Premium framework developed in [1, 2]. This treatise adopts exact ZIRP as its theoretical axiom, while a bounded asymptotic policy rule is treated as a possible implementation regime.

2 Exact ZIRP and aggregate return

Define the inflation risk premium by

$$\pi_i(t) = r_A(t) - r_f(t) - E[i(t)]. \tag{1}$$

Exact ZIRP imposes

$$\boxed{\pi_i(t) = 0 \quad \text{for every admissible } t,} \tag{2}$$

and hence

$$\boxed{r_A(t) = r_f(t) + E[i(t)].} \tag{3}$$

This does not imply that expected inflation is zero or that every other source of uncertainty disappears.

Retain the aggregate-return function specified in the ZIRP framework:

$$r_A(t) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(t-\mu)^2}{2\sigma^2}\right], \qquad \mu \in \mathbb{R}, \quad \sigma > 0. \tag{4}$$

It is positive for every finite t and tends to zero as $t \to \infty$. This is the framework's specified functional form, not a universal empirical law of returns.

For policy implementation, [2] proposes a shrinking envelope

$$|\pi_i(t)| \le \varepsilon(t), \qquad \varepsilon(t) = \varepsilon_0 e^{-\rho t}, \quad \rho > 0. \tag{5}$$

This permits finite-time deviations, unlike exact ZIRP. Existence, uniqueness, and stability are distinct: existence of one exact-ZIRP solution does not prove uniqueness or convergence from every initial condition.

3 Resources and investment dynamics

Use the working resource identity

$$P(t) = S(t) + C(t) + I(t), \tag{6}$$

so that

$$C(t) = P(t) - S(t) - I(t). \tag{7}$$

A complete application must specify whether $I(t)$ denotes an investment flow or an accumulated capital stock and avoid double-counting resources.

The working accumulation relation is

$$I(t+1) = I(t)[1 + r_I(t)]. \tag{8}$$

The investment return $r_I(t)$ must not be identified with the aggregate return $r_A(t)$ without an additional equation.

The preference–ambiguity relation explored here is

$$\alpha = \frac{[1 + \gamma\delta a(t) I(t)][r_I(t) - r_A(t)]}{1 + r_I(t)}, \qquad \alpha \in [0, 1], \tag{9}$$

where $a(t) \ge 0$ is ambiguity, $\gamma \ge 0$ is ambiguity sensitivity, and $\delta > 0$ is an exposure parameter. The preference parameter α is constant. Equation (9) is a working behavioural specification requiring economic justification and empirical scrutiny.

Solving for $r_I(t)$ and substituting into the accumulation relation gives

$$\boxed{I(t+1) = I(t)\frac{[1 + r_A(t)][1 + \gamma\delta a(t) I(t)]}{1 - \alpha + \gamma\delta a(t) I(t)}.} \tag{10}$$

For $0 \le \alpha < 1$, $I(t) > 0$, $a(t) \ge 0$, $\gamma \ge 0$, and $\delta > 0$,

$$I(t+1) - I(t) = I(t)\frac{r_A(t) + \alpha + \gamma\delta a(t) I(t) r_A(t)}{1 - \alpha + \gamma\delta a(t) I(t)} > 0. \tag{11}$$

Thus this recurrence implies strictly increasing investment under these assumptions. A stationary positive investment stock is impossible under this recurrence; this is a property of the chosen equation, not a general law of investment.

At $\alpha = 0$, the ambiguity terms cancel:

$$I(t+1) = I(t)[1 + r_A(t)], \qquad I(t) = I(0)\prod_{s=0}^{t-1}[1 + r_A(s)]. \tag{12}$$

This is a useful benchmark and a model-design question: is ambiguity-independence at $\alpha = 0$ intended, or does the behavioural specification need refinement?

4 A candidate ambiguity function

A parsimonious candidate is

$$a(t) = a_0 e^{-\kappa t}, \qquad a_0 \geq 0, \quad \kappa \geq 0. \tag{13}$$

This is a modelling hypothesis, not a consequence of ZIRP. It allows constant ambiguity when $\kappa = 0$ and decaying ambiguity when $\kappa > 0$.

For $b_0 = \gamma\delta a_0 > 0$, define

$$x(t) = b_0 e^{-\kappa t} I(t), \qquad q = e^{-\kappa}, \qquad c = 1 - \alpha. \tag{14}$$

For $0 < \alpha < 1$, equation (10) becomes

$$x(t+1) = q[1 + r_A(t)]x(t)\frac{1 + x(t)}{c + x(t)}. \tag{15}$$

Since $r_A(t) \to 0$, the limiting autonomous recurrence is

$$x(t+1) = q\, x(t)\frac{1 + x(t)}{c + x(t)}. \tag{16}$$

Its zero fixed point is locally asymptotically stable when

$$\kappa > -\ln(1 - \alpha), \qquad 0 < \alpha < 1. \tag{17}$$

A positive candidate fixed point exists for $0 < \kappa < -\ln(1 - \alpha)$:

$$x_+^* = \frac{e^{-\kappa} - (1 - \alpha)}{1 - e^{-\kappa}}. \tag{18}$$

These are results for the limiting autonomous recurrence, not a convergence proof for the original Gaussian-forced recurrence. The threshold case is nonhyperbolic and requires separate analysis. Convergence of $x(t)$ does not imply convergence of $I(t) = e^{\kappa t}x(t)/b_0$.

5 A theory of economic choice

Let $W(t) = P(t) - S(t)$, so that

$$C(t) + I(t) = W(t), \qquad 0 \leq I(t) \leq W(t). \tag{19}$$

A proposed one-period choice problem is

$$\max_{0 \leq I \leq W} \left\{ u(W - I) + \beta V(I) - \lambda a\, \Psi(I) \right\}, \tag{20}$$

where u is consumption utility, V is investment value, $\beta \geq 0$ weights future benefits, and $\lambda a \Psi(I)$ is a proposed ambiguity adjustment. Its sign and functional form are behavioural hypotheses. Ambiguity differs from ordinary risk with a known probability distribution [6, 8].

For an interior optimum,

$$\boxed{u'(W - I^*) = \beta V'(I^*) - \lambda a\, \Psi'(I^*).} \tag{21}$$

The marginal utility of consumption forgone equals the ambiguity-adjusted marginal benefit of investment.

For illustration, choose

$$u(C) = mC - \frac{h}{2}C^2, \qquad m, h > 0, \tag{22}$$

$$V(I) = RI, \qquad R > 0, \tag{23}$$

$$\Psi(I) = \frac{I^2}{2}. \tag{24}$$

Then the constrained optimum is

$$I^* = \min\left\{W, \max\left\{0, \frac{hW + \beta R - m}{h + \lambda a}\right\}\right\}. \tag{25}$$

When the unconstrained solution is interior and $hW + \beta R - m > 0$, greater ambiguity reduces optimal investment if $\lambda > 0$. This result depends on the illustrative penalty; it is not universal. A dynamic welfare analysis must specify capital transitions, returns, constraints, and discounting consistently [4, 9].

6 Interdisciplinary scope

The following table identifies research interfaces, not claims that the present equations already validate every discipline-specific implication.

Table 1: Research interfaces for Econopsychology

Field	**Contribution and research question**
Mathematics and logic	Existence, admissibility, fixed points, invariance, comparative statics, and convergence proofs.
Economics and welfare economics	Resource constraints, marginal utility, intertemporal choice, distribution, and welfare criteria.
Finance and financial planning	Distinguish capital stock, investment flow, realised return, expected return, and risk premium.
Econometrics and statistics	Identify parameters, test functional forms, quantify uncertainty, and distinguish correlation from causal effects.
Psychology and behavioural science	Model ambiguity attitudes, reference dependence, loss aversion, attention, and bounded rationality [7].
Physics and complex systems	Apply dynamical-systems tools without assuming that economic agents obey physical laws literally.
Biology and neuroscience	Treat adaptive decision-making and bounded attention as hypotheses requiring appropriate evidence.
Chemistry and linguistics	Offer analogies for complex interactions and the meaning of ambiguity; no direct mechanism is assumed here.
Philosophy and religion	Clarify rationality, value, responsibility, and welfare while distinguishing normative commitments from testable claims.
Political science, geopolitics, diplomacy, and international relations	Analyse institutions, credibility, coordination, and external shocks without inferring political conclusions from equations alone.
Warfare and warfare economics	Study resource allocation under conflict only with explicit welfare and safety constraints; no operational military prescription follows.
Computer science, AI/ML, and cybersecurity	Build reproducible simulations, parameter estimation, robust evaluation, secure data handling, and auditable decision support.

7 Empirical strategy and computation

Specify observable proxies for $r_A(t)$, $r_f(t)$, expected inflation, investment, consumption, and ambiguity. Verify that definitions and units are consistent across periods and jurisdictions. Estimate parameters only after determining which observations distinguish competing mechanisms.

The exact-ZIRP restriction is

$$r_A(t) - r_f(t) - E[i(t)] = 0. \tag{26}$$

Empirical implementation should report measurement error in each component. A bounded implementation can test whether the measured premium remains inside equation (5). Neither a fitted curve nor a near-zero estimate proves the axiom; it provides evidence relevant to the model.

A minimal simulation algorithm is:

Input: $\mu, \sigma, \alpha, \gamma, \delta, a_0, \kappa, I(0), T$.
For $t = 0, \ldots, T-1$:

$$\begin{aligned} &r_A(t) \leftarrow \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(t-\mu)^2}{2\sigma^2}\right], \\ &a(t) \leftarrow a_0 e^{-\kappa t}, \\ &I(t+1) \leftarrow I(t)\frac{[1+r_A(t)][1+\gamma\delta a(t)I(t)]}{1-\alpha+\gamma\delta a(t)I(t)}. \end{aligned} \tag{27}$$

Record the state and check all domain restrictions.
Output: trajectories, constraints, and sensitivity results.

The implementation must reject parameter combinations that make a denominator zero or violate the declared domain. It should test limiting cases such as $\alpha = 0$. Simulation can reveal counterexamples and numerical patterns but cannot substitute for proof.

8 Exact-ZIRP existence criterion

Let $\mathcal{F}$ denote admissible configurations satisfying the resource, institutional, and behavioural restrictions, and define

$$\mathcal{Z} = \{z \in \mathcal{F} : \pi_i(t; z) = 0 \text{ for every admissible } t\}. \tag{28}$$

Proposition. An admissible exact-ZIRP configuration exists if and only if

$$\boxed{\mathcal{Z} \neq \varnothing.} \tag{29}$$

Proof. By definition, any element of $\mathcal{Z}$ satisfies admissibility and exact ZIRP. Thus nonemptiness supplies an admissible solution. Conversely, any admissible exact-ZIRP solution belongs to $\mathcal{Z}$, so the set is nonempty. □

This proposition formalises the existence criterion but does not prove nonemptiness for a particular calibrated economy. That requires a model-specific existence argument. Existence, uniqueness, stability, and empirical fit remain distinct claims.

9 Schematic structure

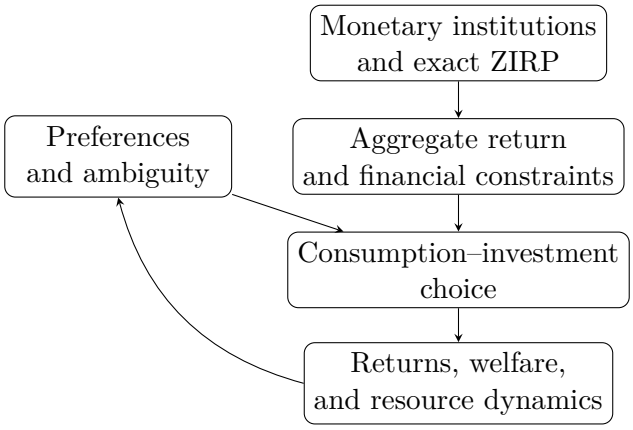


Figure 1: Conceptual relationships proposed for Econopsychology; arrows are hypotheses, not established causal effects.

10 Limitations and research agenda

1. Derive or justify the preference–ambiguity relation in equation (9).
2. Distinguish investment flows from capital stocks and ensure coherent accounting.
3. Prove convergence results for the original nonautonomous recurrence.
4. Establish model-specific existence of exact-ZIRP equilibria; the proposition above is a criterion, not a substantive existence proof.
5. Specify the relationship between $r_I(t)$ and $r_A(t)$.
6. Identify ambiguity empirically and test its functional form.
7. Compare welfare under alternative allocations, including distribution and uncertainty.
8. Publish code, data provenance, diagnostics, and robustness checks for reproducibility.

11 Conclusion

Econopsychology is proposed as a theory of economic choice under preferences, ambiguity, financial constraints, and exact ZIRP. The Gaussian return function and working investment recurrence provide a dynamical starting point. The resource constraint and constrained optimisation problem shift attention from investment growth alone to the trade-off between present consumption and future benefits.

The recurrence implies strictly increasing investment under stated assumptions, ambiguity cancels when $\alpha = 0$, and the limiting transformed recurrence has candidate stability regimes. These results do not establish that indefinite accumulation is optimal, that every trajectory converges, or that an exact-ZIRP solution exists for every economy. The governing principle is that investment is a means to future benefits, while economic choice determines whether those benefits justify the resources committed today.

References

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Glossary

Ambiguity Uncertainty about probabilities, models, or distributions relevant to a choice.

Aggregate return, $r_A(t)$ The return function specified by the ZIRP framework.

Exact ZIRP The condition $\pi_i(t) = 0$ for every admissible time.

Inflation risk premium, $\pi_i(t)$ The difference $r_A(t) - r_f(t) - E[i(t)]$.

Investment stock Capital accumulated for future production or returns; distinct from an investment flow.

Investment return, $r_I(t)$ Return attributed to investment, distinct from $r_A(t)$ unless related by an explicit equation.

Preference parameter, α The constant parameter in the working preference–ambiguity relation, with $\alpha \in [0, 1]$.

Resource constraint The condition defining feasible allocations between consumption and investment.

ZIRP envelope A shrinking bound that permits finite-time deviations from exact ZIRP.

Welfare A criterion for evaluating outcomes, including consumption, uncertainty, and distribution.

The End